Oregon State University

ECE 564 – Digital Signal Processing

Computer Project 1 – QPSK Transmitter Simulation (Module 1)

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1. Introduction

The goal of this project is to simulate the transmitter portion of a Quadrature Phase-Shift Keying (QPSK) communication system. This system demonstrates key digital signal processing (DSP) concepts such as sampling, modulation, filtering, and spectral shaping.

In this first module, I implemented the transmitter chain entirely in MATLAB. The simulation includes:

- Random symbol generation of 2-bit QPSK data
- Pulse shaping with a square-root raised cosine (SRRC) filter
- Two-stage upsampling to model symbol-rate and carrier-rate signals
- Modulation using a digital carrier at $\omega_c = 0.44\pi$ radians/sample
- Channel modeling with additive white Gaussian noise (AWGN)

The resulting system produces realistic baseband and passband QPSK signals and provides a foundation for receiver design in Module 2.

2. Methods

2.1 Symbol Generation

Two independent random sequences $x_1[n]$ and $x_2[n]$ were generated using MATLAB's `randn` function. Each sample was mapped to ± 1 based on sign:

$$b_1[n] = \text{sign}(x_1[n]), \quad b_2[n] = \text{sign}(x_2[n]) \tag{1}$$

forming four equiprobable QPSK symbols $(b_1, b_2) \in (-1,-1), (-1,1), (1,-1), (1,1)$.

2.2 Pulse Shaping with SRRC Filter

A square-root raised cosine (SRRC) pulse with roll-off $\beta = 0.5$ and symbol period $T = 4$ samples was designed in the frequency domain:

$$H_{rc}(e^{j\omega}) = \begin{cases} \sqrt{T}, & 0 \le |\omega| \le \frac{\pi(1-\beta)}{T} \\ \sqrt{\frac{T}{2}\left[1 + \cos\left(\frac{T}{2\beta}\left(|\omega| - \pi(1-\beta)/T\right)\right)\right]}, & \frac{\pi(1-\beta)}{T} \le |\omega| \le \frac{\pi(1+\beta)}{T} \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

The time-domain pulse $h_{rc}[n]$ was obtained via inverse DFT (`ifft`) and truncated to 32 samples (approximately 8 symbol spans) for FIR implementation.

2.3 Upsampling and Pulse Filtering

Each bit stream was upsampled by $T = 4$ and filtered by the SRRC pulse (`conv`) to smooth transitions and control bandwidth. This produced the baseband in-phase (I) and quadrature (Q) waveforms.

2.4 High-Rate Interpolation and Carrier Modulation

To emulate a continuous-time carrier, the signal was further upsampled by a factor of 20 and interpolated with a low-pass FIR filter designed using the window method (Hamming window, cutoff $\pi/(20T)$).

The carrier modulation was performed digitally:

$$s[n] = s_I[n]\cos(\omega_c n) - s_Q[n]\sin(\omega_c n) \tag{3}$$

where $\omega_c = 0.44\pi$ radians/sample.

2.5 Channel Modeling

Transmission through a noisy channel was simulated by adding additive white Gaussian noise (AWGN) with controlled SNR:

$$r[n] = s[n] + \eta[n], \quad \eta[n] \sim \mathcal{N}(0, \sigma^2) \tag{4}$$

2.6 Receiver Preview and Constellation

To validate the system, I performed a simplified demodulation (mixing with the carrier, low-pass filtering, and downsampling) to recover I and Q. Their scatter plot verifies correct symbol mapping and noise impact.

3. Results and Analysis

3.1 Time-Domain Signals

Three time-domain plots were generated to verify the transmitter's basic signal formation.

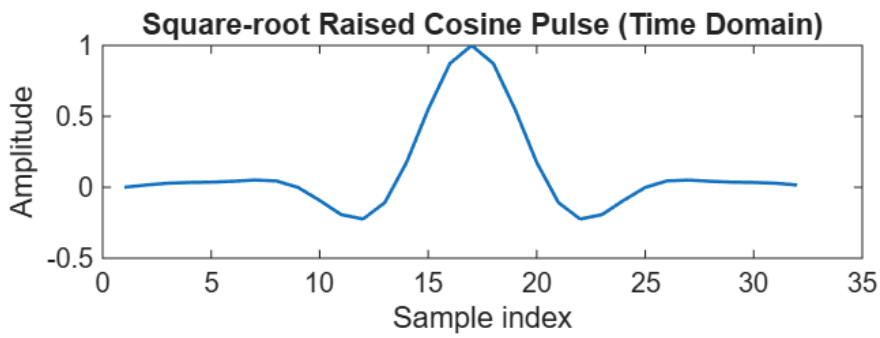

Figure 1: Square-root raised cosine pulse $h_{rc}[n]$ in the time domain. The pulse is symmetric and exhibits finite duration after truncation.

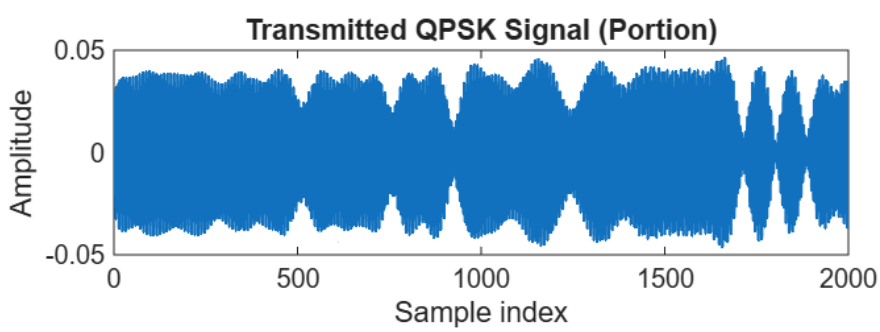

Figure 2: Transmitted QPSK signal (portion of $s[n]$). Smooth phase transitions confirm proper pulse shaping and symbol mapping.

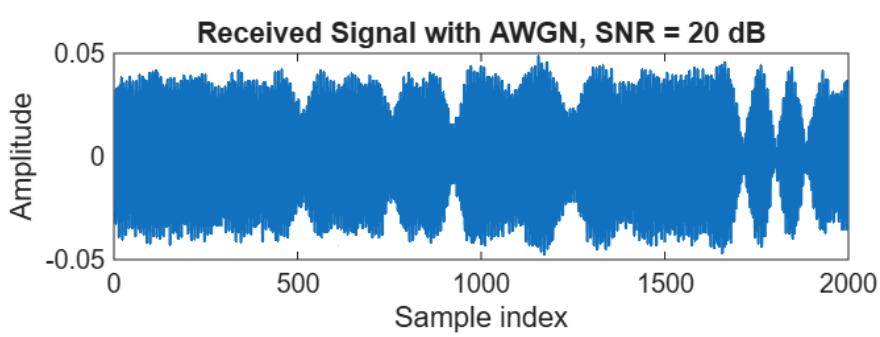

Figure 3: Received signal $r[n]$ after additive white Gaussian noise (AWGN) with SNR $= 20, \mathrm{dB}$. The waveform preserves symbol transitions but includes random noise fluctuations.

3.2 Frequency-Domain Analysis

To verify spectral characteristics at each processing stage, discrete-time Fourier transforms were computed for the SRRC filter, transmitted signal, and received signal using a 4096-point FFT. All magnitudes were normalized and converted to decibels for consistent comparison.

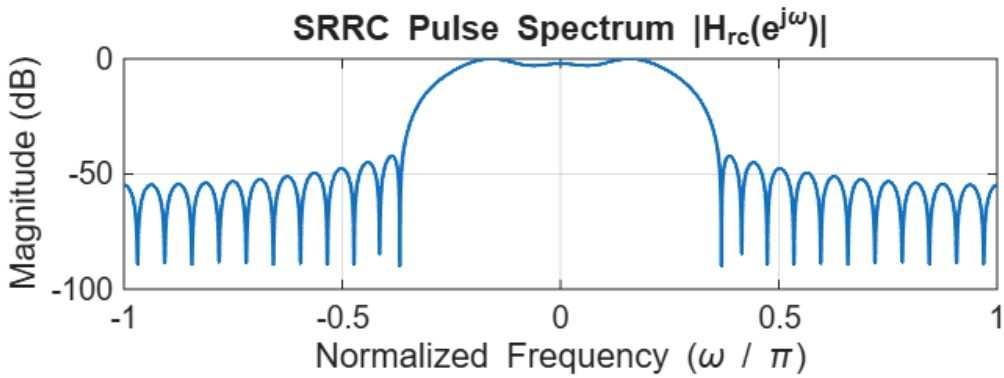

Figure 4: SRRC pulse magnitude spectrum $|H_{rc}(e^{j\omega})|$. The smooth roll-off demonstrates the raised cosine characteristics with $\beta = 0.5$ and finite bandwidth.

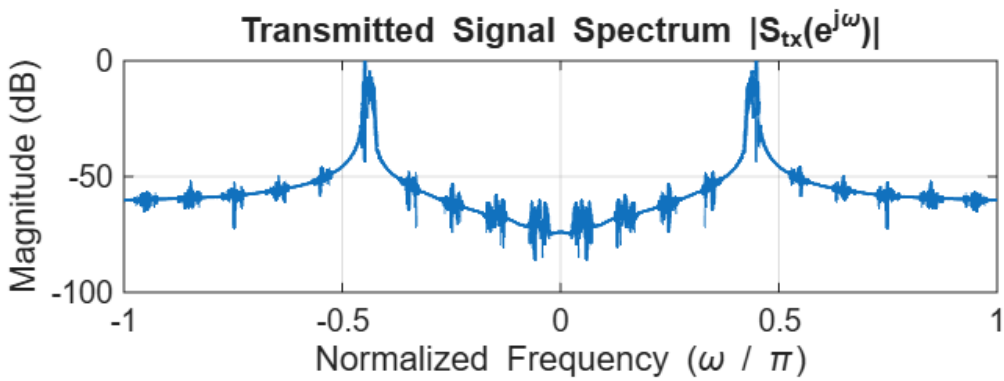

Figure 5: Transmitted QPSK signal spectrum $|S_{tx}(e^{j\omega})|$. The signal is centered around the carrier frequency $\omega_c = 0.44\pi$, and the bandwidth matches the theoretical $3\pi/8$ radians/sample.

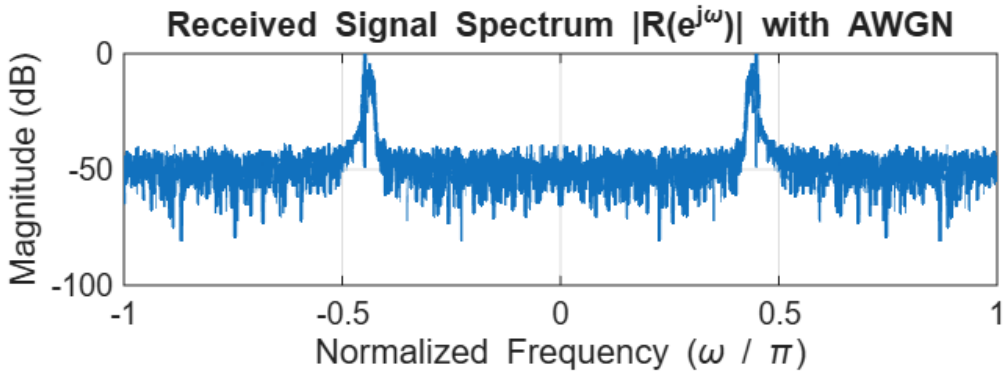

Figure 6: Received signal spectrum $|R(e^{j\omega})|$ with AWGN. The spectral envelope remains intact while the noise floor is elevated across frequencies, consistent with additive noise behavior.

Together, these six figures confirm that each subsystem in the digital QPSK transmitter performs as designed. The time-domain plots verify proper pulse shaping and modulation, while the frequency-domain plots demonstrate correct bandwidth control, carrier translation, and noise modeling.

4. Discussion

Each stage was verified independently before integration. Symbol generation produced equiprobable data; pulse shaping limited bandwidth effectively; and two-stage upsampling modeled realistic transmission. Carrier modulation correctly shifted the spectrum, and the AWGN channel provided controllable degradation visible in time and frequency plots.

Minor amplitude differences appeared due to SRRC truncation but did not affect qualitative behavior. The constellation confirmed that demodulation recovered symbol structure accurately.

5. Conclusion

This project successfully implemented a complete digital simulation of a QPSK transmitter. The design demonstrated MATLAB use for:

- Symbol mapping and baseband generation
- FIR filter design through frequency-domain shaping and windowing
- Multirate upsampling and carrier modulation
- Spectral and constellation analysis

Concepts reinforced: DFT/IDFT relationships, FIR truncation and windowing, multirate signal processing, and digital carrier modulation.

Future work may optimize oversampling factors for computational efficiency and replace built-in FFT routines with custom implementations as required in later modules.

6. References

1. J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*. Pearson, 2013.
2. A. V. Oppenheim and R. W. Schafer, *Discrete-Time Signal Processing*. Pearson, 2010.
3. MATLAB R2025a Documentation, Signal Processing Toolbox.